

Mathematical Problems

CSE4261: Neural Network and Deep Learning
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YOLO

1. Consider a simplified YOLO model that predicts one bounding box for a grid cell. For a particular training example:

Ground Truth

- Object exists: $P_c = 1$
- Bounding box:
 - $x = 0.6$; $y = 0.7$; $w = 0.5$; $h = 0.4$
 - Class label: Cat $\rightarrow$ one-hot vector = $[1, 0]$

Model Prediction

- Predicted objectness: $\hat{P}_c = 0.8$
- Predicted box: $\hat{x} = 0.5$; $\hat{y} = 0.6$; $\hat{w} = 0.4$; $\hat{h} = 0.5$
- Predicted class probabilities: Cat = 0.7; Dog = 0.3

Assume:

- Localization loss weight: $\lambda_{coord} = 5$

Using the simplified YOLO loss, compute the total loss.

2. An object detection model predicts the following bounding boxes for the same class ("cat"):

Box	Coordinates (x1, y1, x2, y2)	Confidence Score
A	(1,1,5,5)	0.95
B	(2,2,6,6)	0.90
C	(7,7,10,10)	0.85

Assume:

- IoU threshold = 0.30
- Higher confidence boxes are selected first.

Answer the following:

- A. Compute the IoU between Box A and Box B.
- B. Determine whether Box B will be suppressed.
- C. Compute the IoU between Box A and Box C.
- D. Apply Non-Maximum Suppression step-by-step and determine the final boxes retained.

eXplainable AI (XAI)

1. Assume a CNN-based 3-class classifier which can be used for CAM.
Let a 5x5 image be passed through the assumed CNN-based 3-class classifier.

$$I = \begin{bmatrix} 1 & 2 & 3 & 2 & 1 \\ 0 & 1 & 2 & 1 & 0 \\ 1 & 3 & 4 & 3 & 1 \\ 0 & 1 & 2 & 1 & 0 \\ 1 & 2 & 3 & 2 & 1 \end{bmatrix}$$

Let the last convolutional layer of a CNN has three feature maps (size 2x2)

$$f_1 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad f_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad f_3 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

Let the fully connected layer weights (for class c)

$$w^c = [2, -1, 1]$$

- a. Compute Global Average Pooling (GAP) outputs F_1, F_2, F_3 .
- b. Compute the class score:

$$S_c = \sum_k w_k^c F_k$$

- c. Compute the Class Activation Map:

$$M_c(x, y) = \sum_k w_k^c f_k(x, y)$$

- d. Normalize the CAM:

$$M' = \frac{M - \min(M)}{\max(M) - \min(M)}$$

- e. Explain how this 2x2 CAM would be mapped back to the 5x5 input image.

2. Show steps of CAM for your own three sets of $\{I, f_1, f_2, f_3, w^c\}$.
3. Show steps of Grad-CAM for the above network for the above image.
4. Show Grad-CAM when multiple FC layers exist.
5. Visualize before vs after ReLU as heatmaps
6. Show what happens if we DON'T use ReLU (full attribution).
7. Say model $F(x_1, x_2) = x_1^2 + 2x_2$, input $x = (2, 3)$, baseline $x' = (0, 0)$, $m = 10$. compute the Integrated Gradients (IG) using a discrete approximation. Show all steps as in a real implementation of the IG algorithm.

8. Show all steps as in a real implementation of the IG algorithm for your own three sets of $\{F, x, x', m\}$.
9. Compare Grad-CAM vs Integrated Gradients (IG) using the same situation.

Convolutional Neural Network

1. Consider the following grayscale input image I of size 5×5 :

$$I = \begin{bmatrix} 1 & 2 & 0 & 1 & 3 \\ 4 & 1 & 2 & 0 & 1 \\ 1 & 3 & 1 & 2 & 2 \\ 0 & 2 & 4 & 1 & 0 \\ 1 & 1 & 2 & 3 & 1 \end{bmatrix}$$

A convolution layer uses the following 3×3 kernel:

$$K = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

- A. Calculate the output size after performing the convolution operation with Stride = 1 and Padding = 0.
 - B. Compute every element of the convolution output feature map. Show all intermediate calculations.
 - C. Compute the final pooled feature map and its size after applying a 2×2 max-pooling operation with stride 2.
 - D. Calculate the total trainable parameters of the convolution layer, including biases when Input channels = 1, Number of filters = 10, and Kernel size = 3×3 .
 - E. If padding is changed from $P = 0$ to $P = 1$:
 - I. What will be the new output size?
 - II. Explain why the output size changes.
 - F. How many multiplications are required to compute one output pixel for one filter?
 - G. How many multiplications are required to compute one output pixel for one filter?
 - H. How many multiplications are required if there are 10 filters?
2. Assume the input feature map has size $4 \times 4 \times 2$ (Height $\times$ Width $\times$ Channels):
 - 3.

$$X_1 = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix} \quad X_2 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 \end{bmatrix}$$

Also assume a Convolution Kernel ($2 \times 2 \times 2$):

$$K_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad K_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Perform the following operations and compute the resulting feature map size (and values where necessary):

- A. Regular convolution without zero padding and with stride = 2
- B. Regular convolution with zero padding and with stride = 2
- C. Depthwise convolution without zero padding and with stride = 1

Clearly show all steps, including:

- a. Output dimension calculations
 - b. Intermediate computations
 - c. Final feature maps
4. Assume the convolution output contains the value -5 . Compute the output when using: ReLU, Sigmoid, and Tanh.